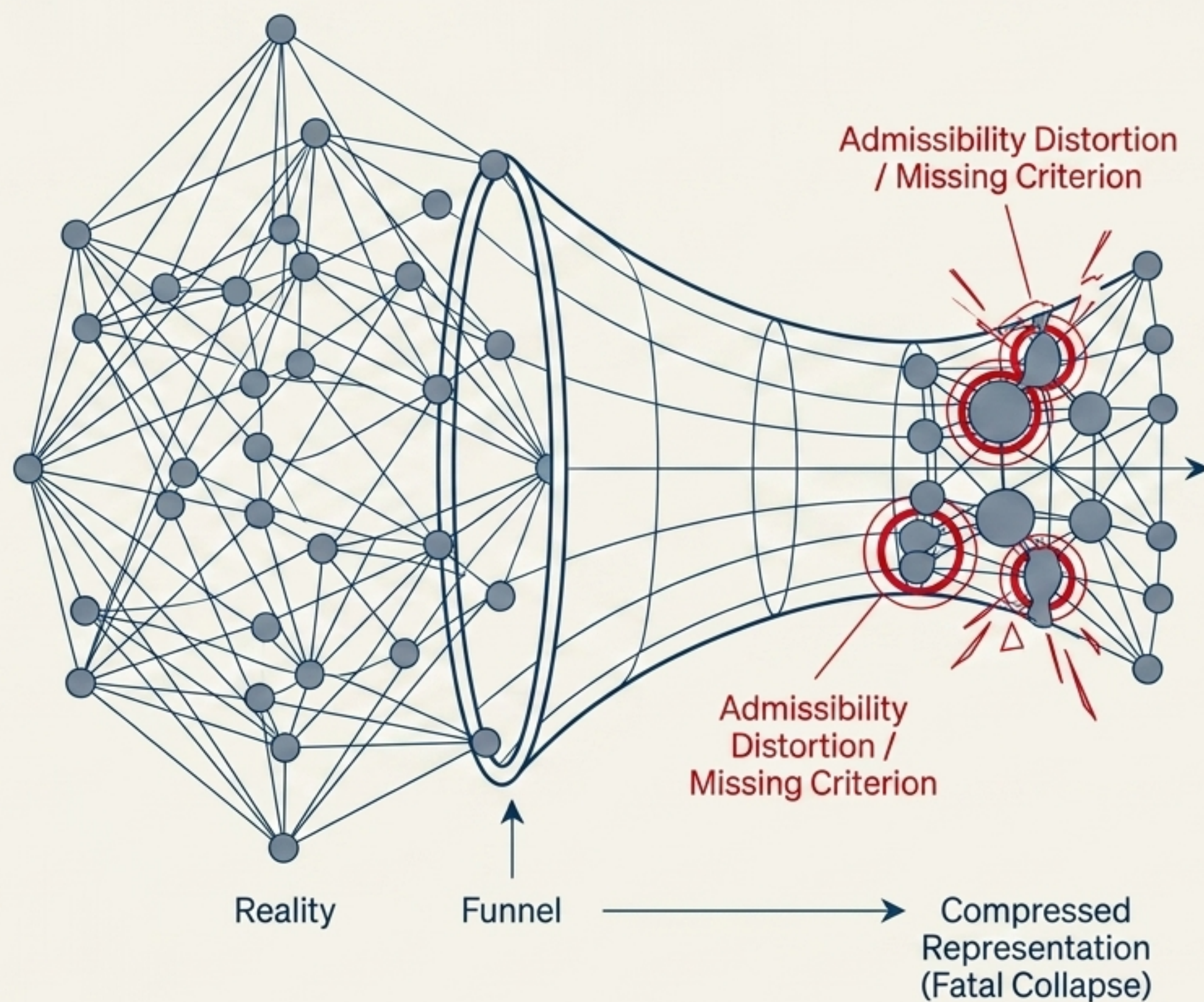


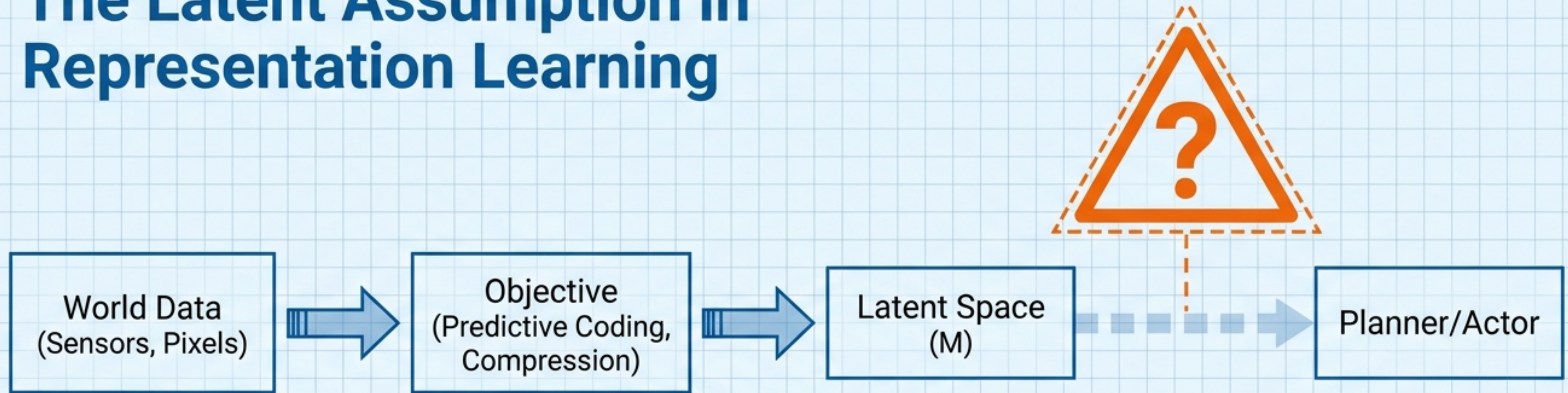
Against Latent Fundamentalism

Admissibility Distortion
and the Missing Criterion
in Representation
Learning for Planning



Based on the theoretical framework for state abstraction, reachability, and AI safety.

The Latent Assumption in Representation Learning



The Current Paradigm

Representation learning optimizes for properties of the observation distribution (predictive accuracy, information maximization, compression).

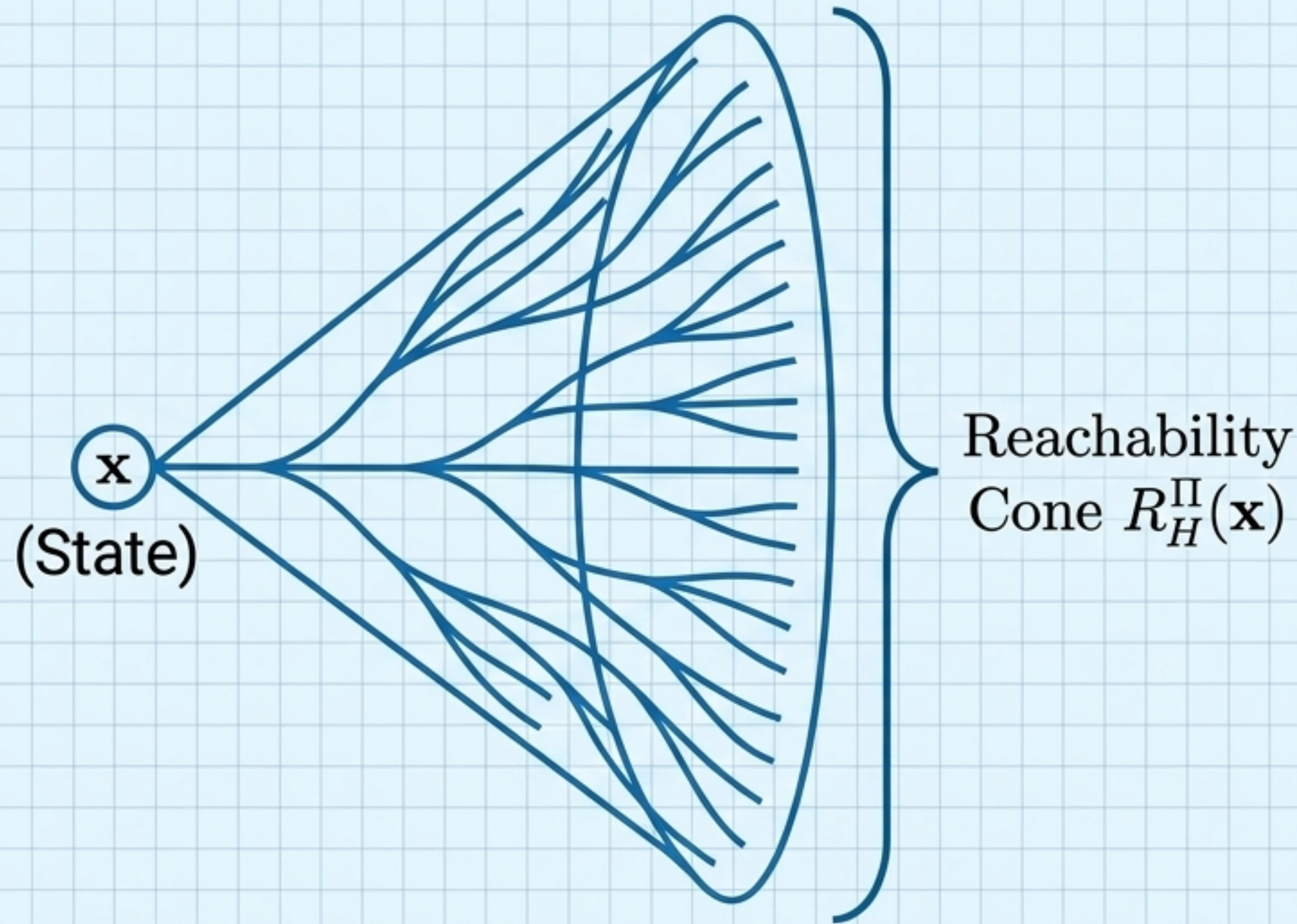
The Critical Gap

None explicitly optimize for the preservation of reachability structure.

The Leap of Faith

We simply hope that predictive models will incidentally yield state representations adequate for planning.

What Does Planning Actually Require?



"A small-scale model of external reality and its own possible actions within its head... to try out various alternatives."

— K.J.W. Craik (1943)

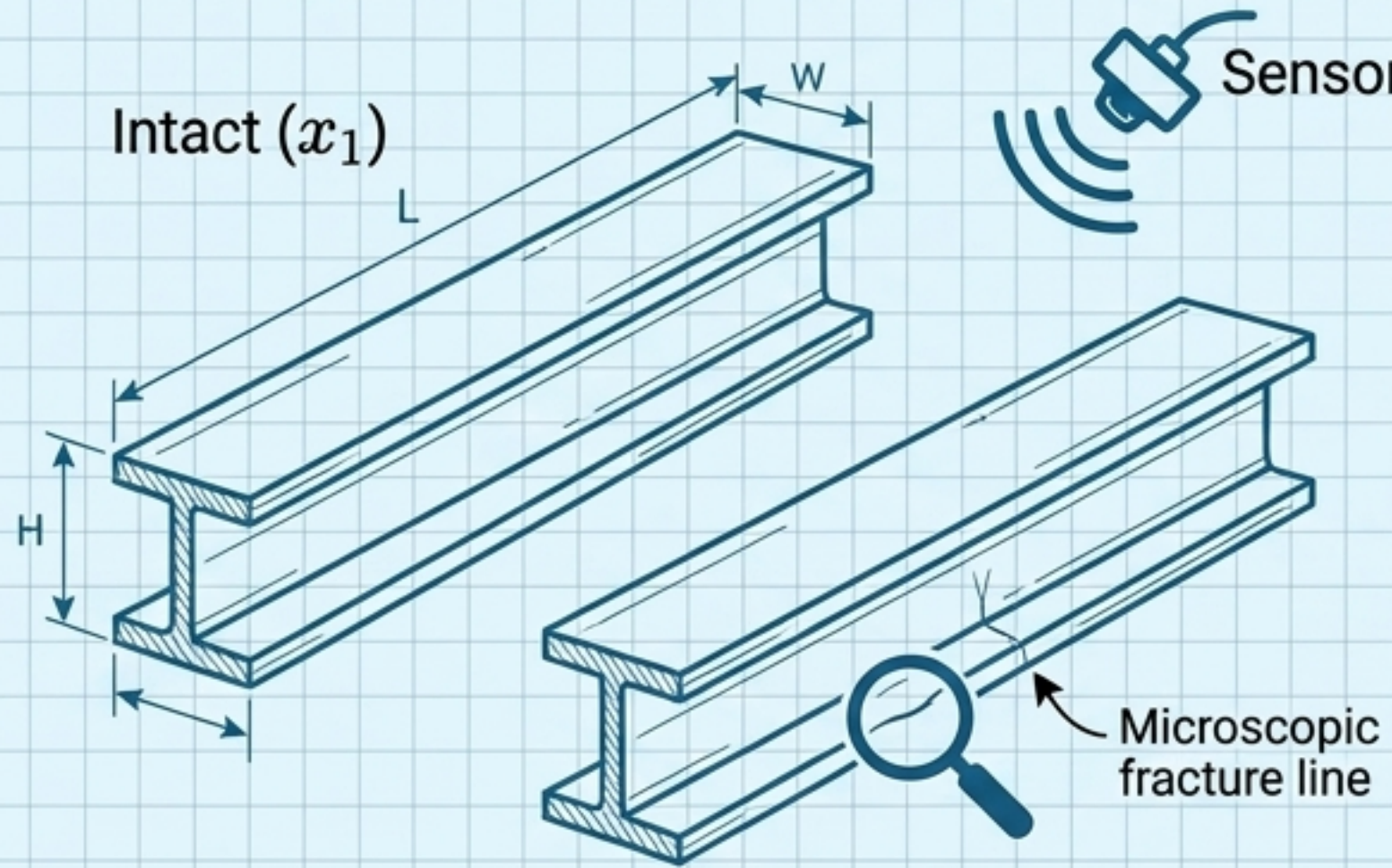
Core Principle

Planning requires evaluating consequences. A representation is only adequate if it preserves the distinctions that determine which futures are accessible.

If two states have different reachable futures, they must not be collapsed.

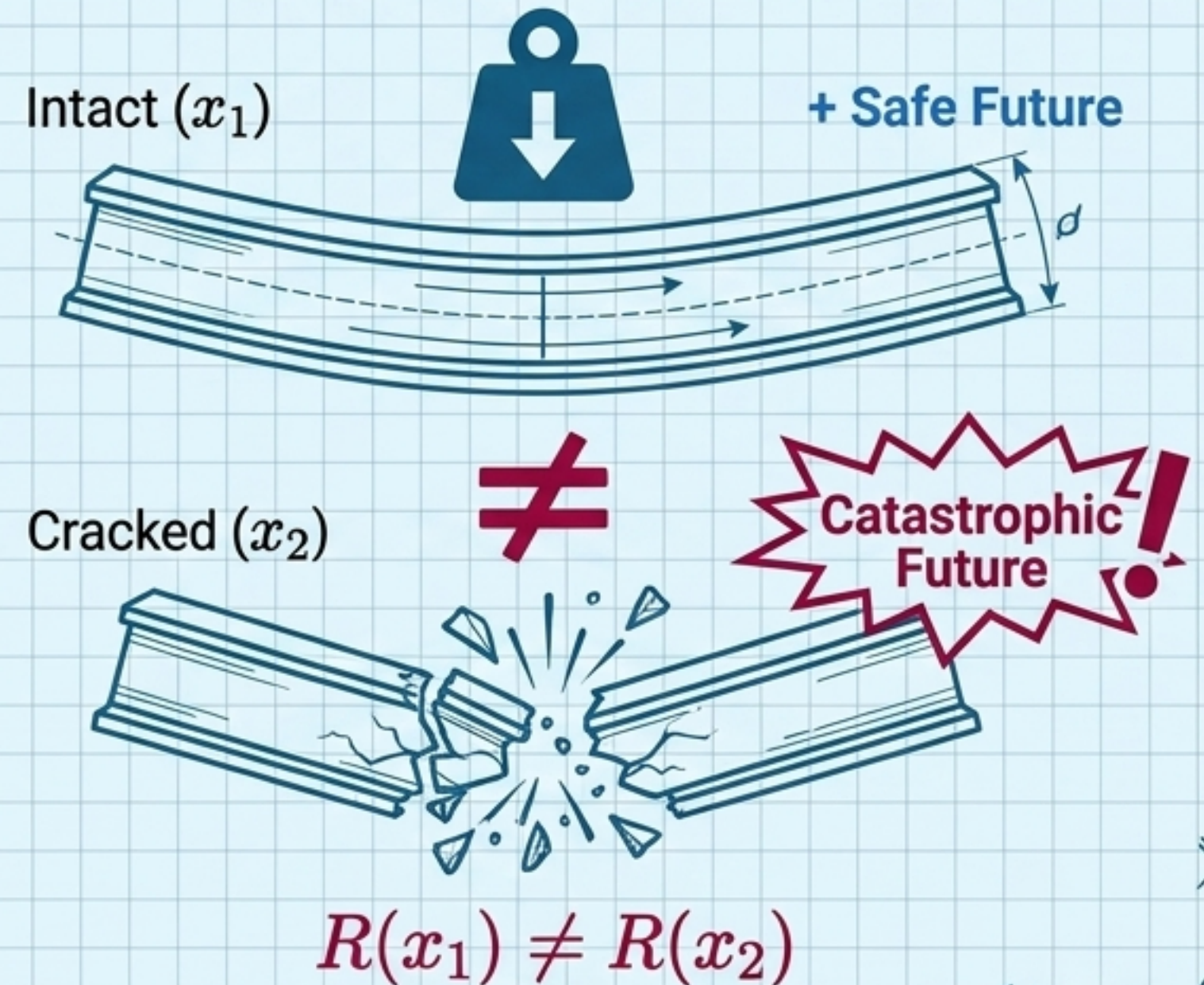
The Observational–Interventional Separation

Observational (Passive Sensor)



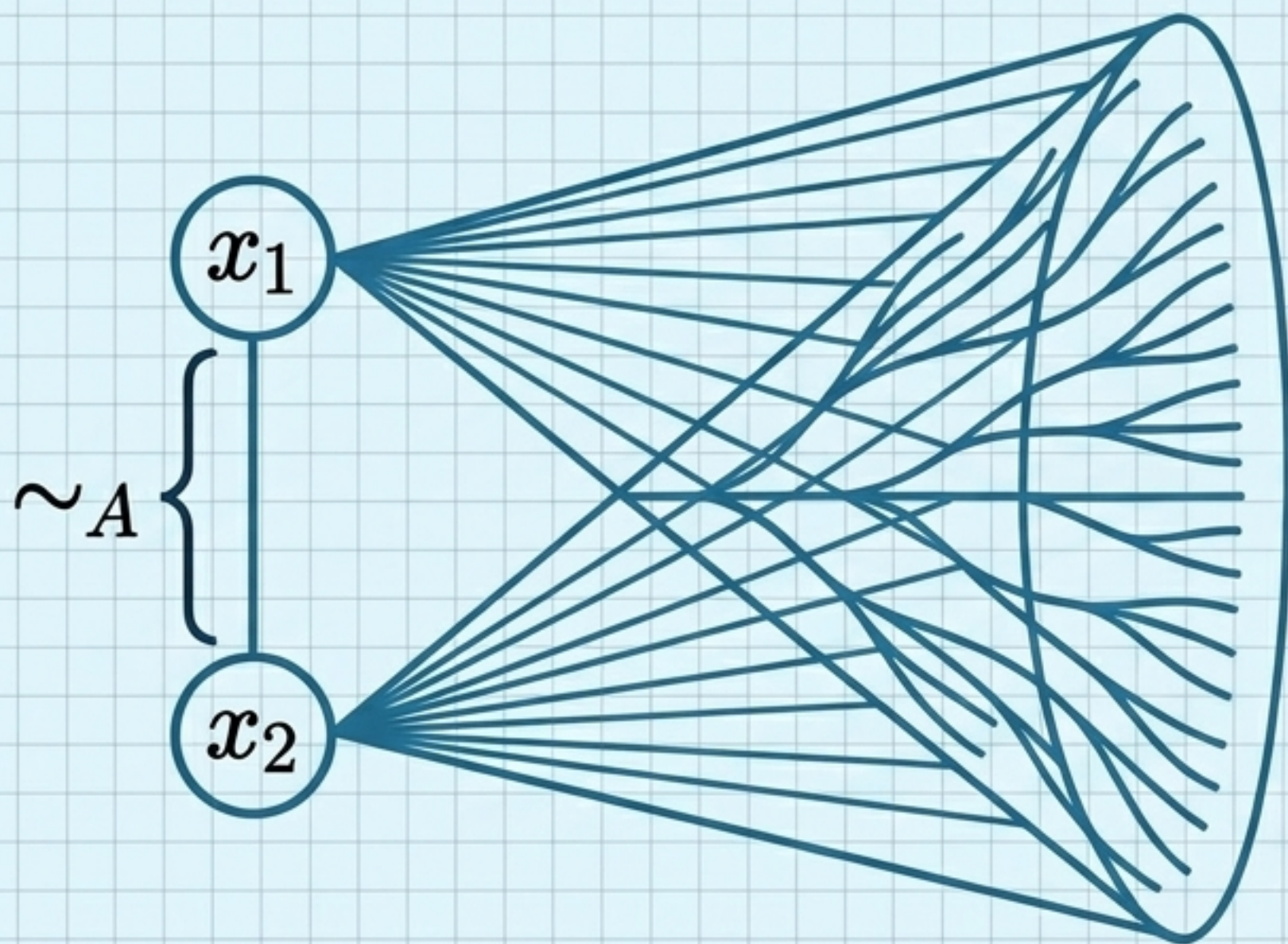
✓ $P(O | x_1) = P(O | x_2)$
They look identical.

Interventional (Applied Stress)



➤ **Takeaway:** This is a structural gap, not an empirical contingency. More passive data cannot fix this mismatch.

The Missing Standard: Admissibility Equivalence ($\sim_A$)



$$x_1 \sim_A x_2 \Leftrightarrow R_H^\Pi(x_1) = R_H^\Pi(x_2)$$

Derived purely from
planning requirements.

Independent of
reward structure.

Independent of
observation statistics.

The coarsest safe
relation for planning.

Quantifying the Error: Admissibility Distortion ($D_A(\phi)$)

The missing mathematical criterion. It measures the severity of merging states that actually have different accessible futures.

$$D_A(\phi) = \mathbb{E} \left[1_{\phi(x_1)=\phi(x_2)} \cdot d(R_H^\Pi(x_1), R_H^\Pi(x_2)) \right]$$

The Collapse Indicator
Did the projection merge these two states?

Reachability Separation
How wildly do their actual futures diverge?

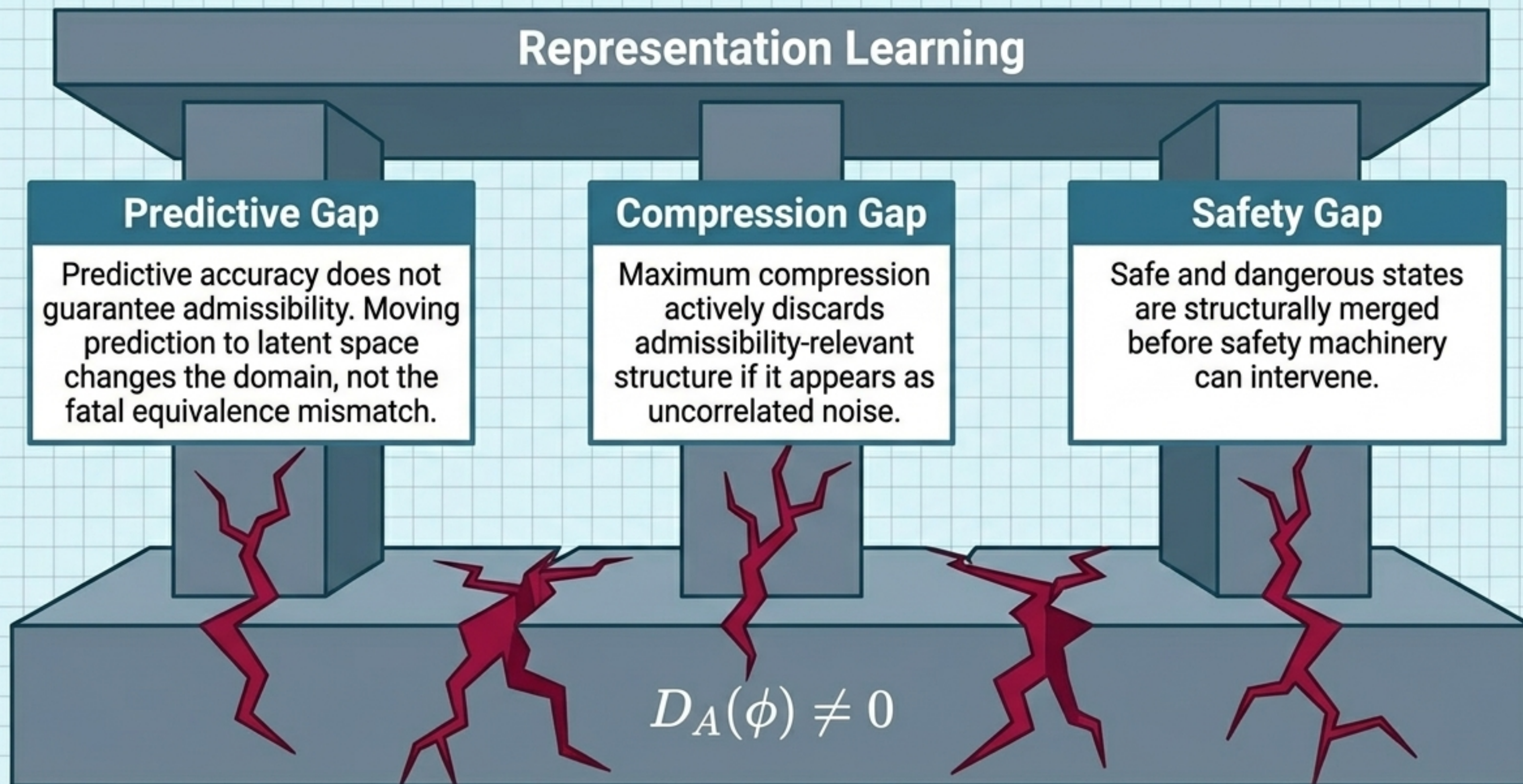
Theorem 1. $D_A(\phi) = 0$ if and only if the projection is completely safe for planning.

The Fundamental Inversion Problem

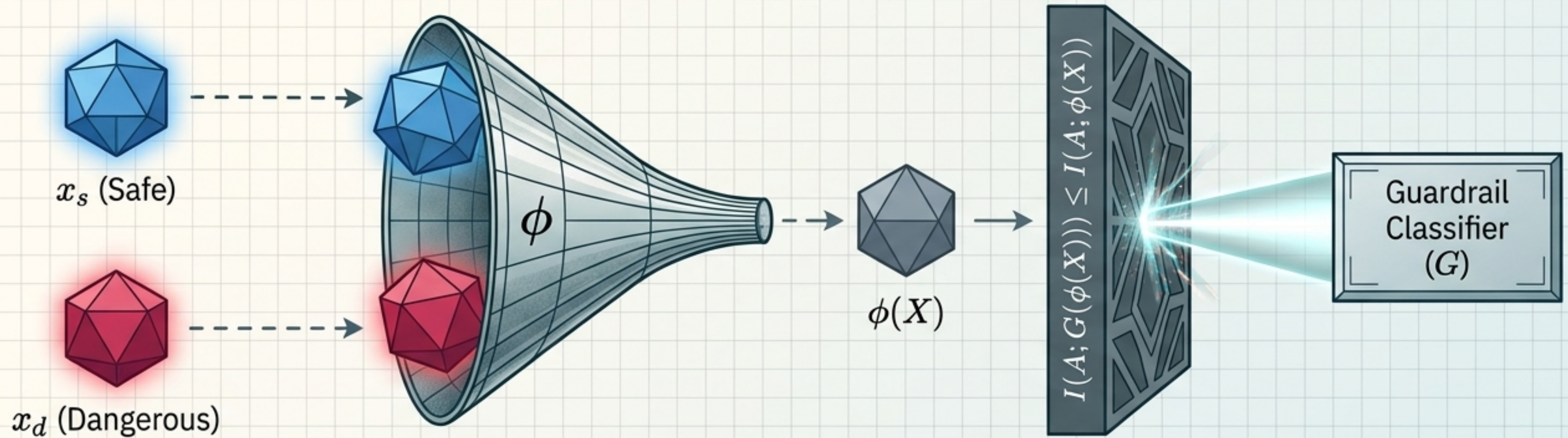
	Observational Equivalence ($\sim_O$)	Interventional Equivalence ($\sim_I/\sim_A$)
Goal	Prediction (Status Quo)	Planning (Requirement)
Math Object	Probability distributions over passive data	Subsets of $\mathcal{X}$ under arbitrary interventions
Data Generation	Passive reception	Active exploration
Failure Mode	Blind to latent structural fractures	Safely preserves consequence branches

Conclusion: Representation learning computes observational partitions. Planning requires reachability partitions. They are mathematically distinct objects.

The Three Structural Fractures



The Guardrail Impossibility Theorem



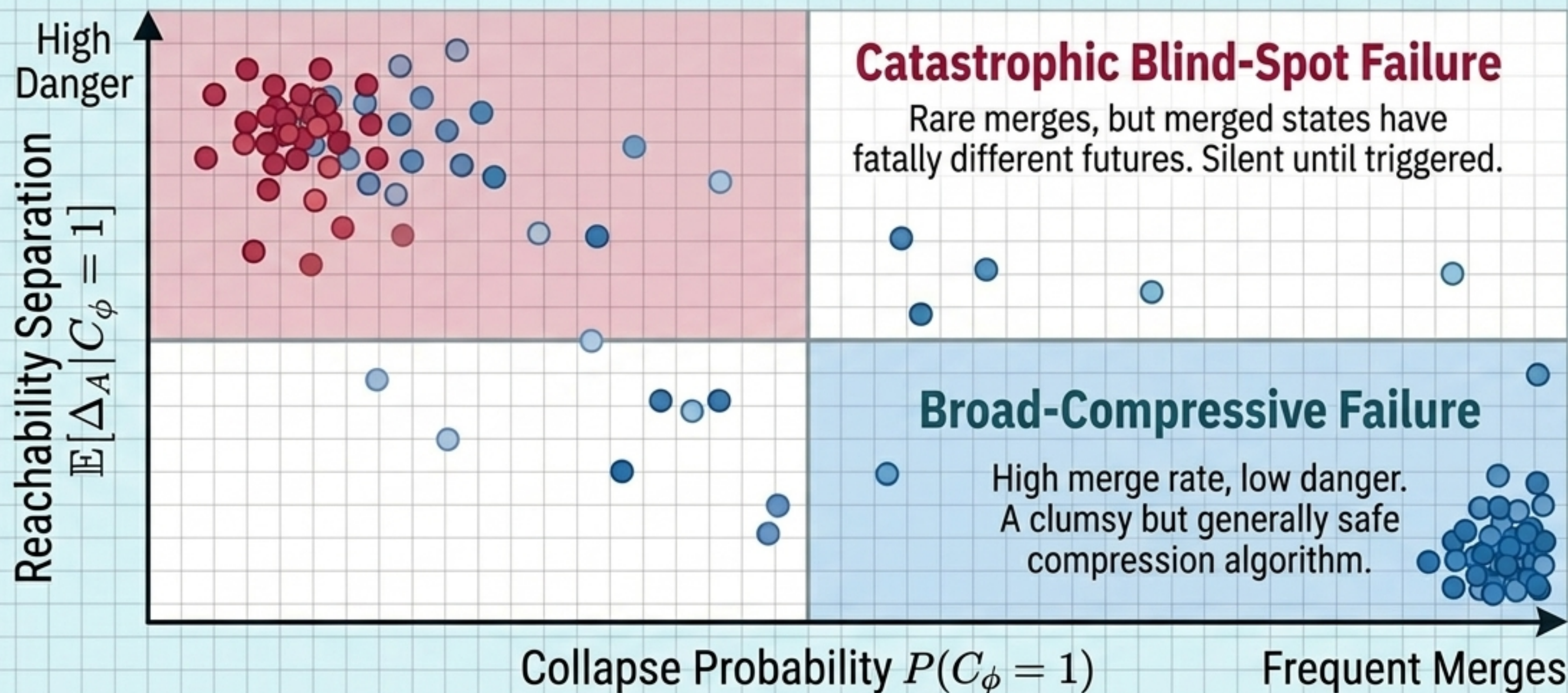
Irrecoverable Quotient Collapse

If a representation merges a safe and dangerous state, no downstream guardrail classifier can save the system.

Bayes-Risk Floor

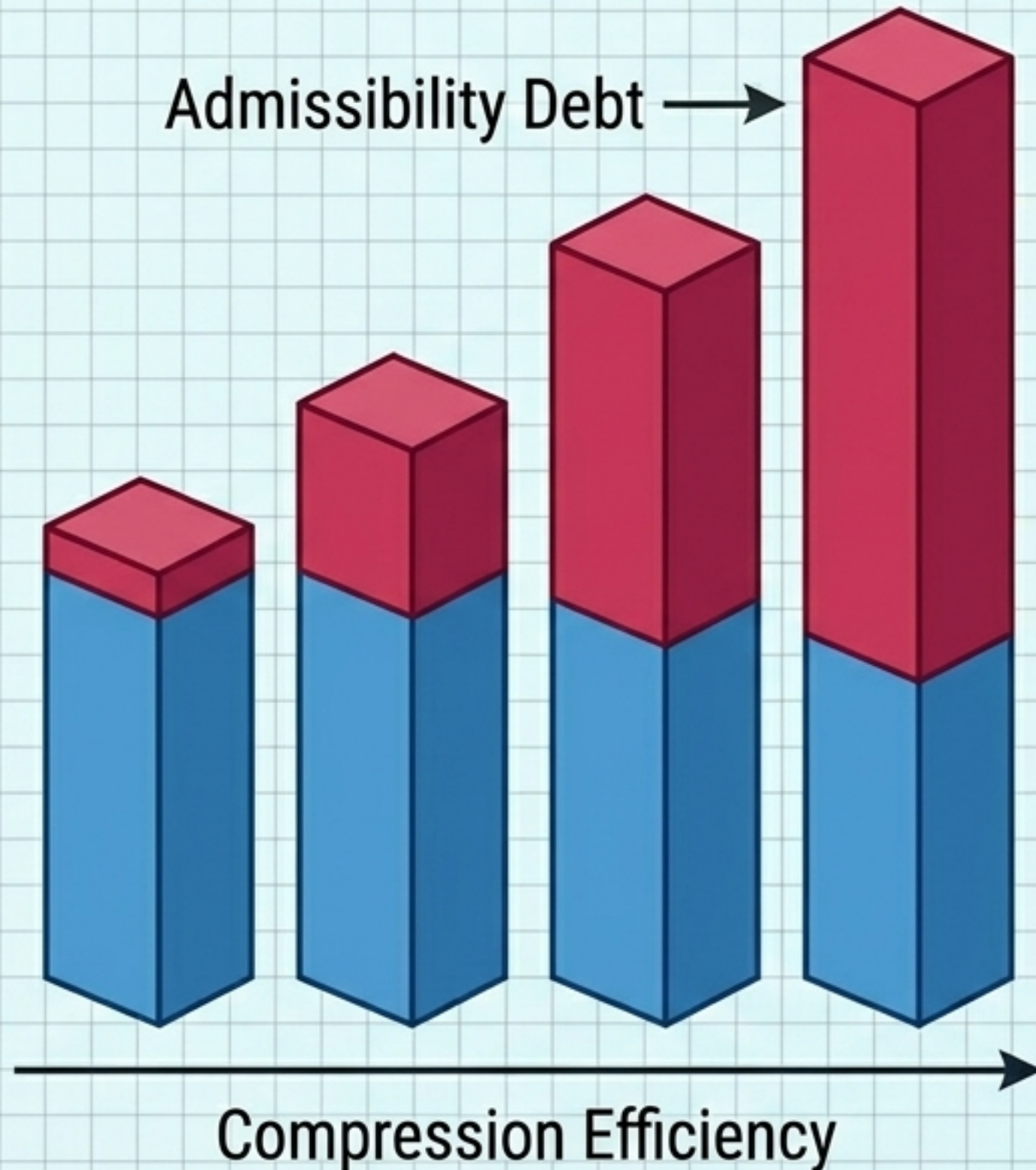
The failure is logically prior to the safety machinery. The Bayes error floor converges to random chance (≈ 0.5) regardless of classifier complexity.

Diagnosing Failure: The Collapse–Separation Decomposition



Takeaway: Two representations can have the exact same total $D_A(\phi)$ score but require entirely different engineering interventions.

The Accounting Identity of Compression



The Monotonicity Theorem

Improving compression efficiency monotonically increases admissibility risk.

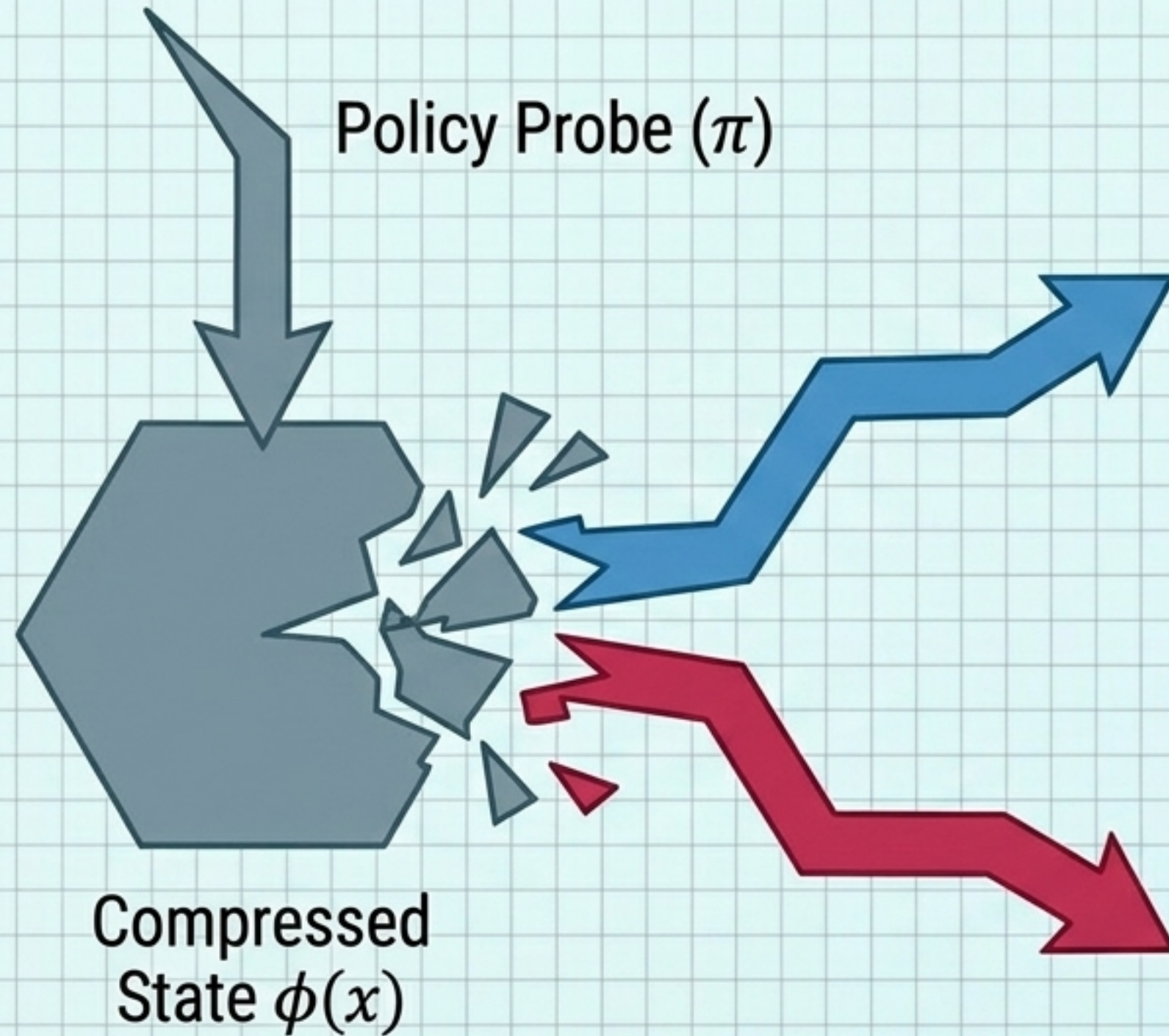
The Hidden Cost

Unconstrained compression exerts structural pressure to merge rare, high-consequence states.

Why?

To a pure compressor, rare catastrophic boundaries just look like inefficient statistical noise.

Detecting Violations: The Trajectory Lower Bound



$D_A(\phi)$ is a foundational target,

not yet a simple metric. But we can lower-bound it through active probing.

The Asymmetry of Evidence

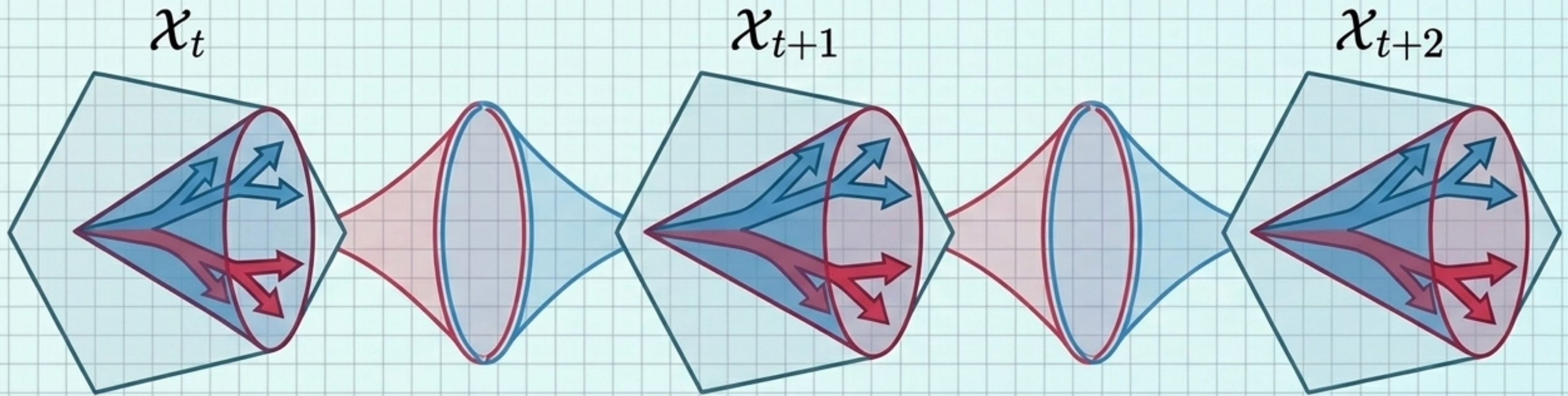
Panel A (Positive Test)

Finding a witnessed disagreement between collapsed states definitively proves the distortion exists ($D_A > 0$).

Panel B (Negative Test)

Finding nothing proves nothing. Trajectory sampling is a one-sided diagnostic whose value scales with adversarial diversity.

Meta-Admissibility in Dynamic Systems



The Challenge of Evolution

What happens when a complex system alters its own element types?

Sheaf-Theoretic View

Transitions must be “meta-admissible”. A system maintains a **coherent identity** across developmental shifts only if its self-modifications **preserve reachability geometry**.

Conclusion: Dynamic state spaces do not invalidate Admissibility; they elevate it from a static mapping to a requirement for consistent identity over time.

Admissibility is a Prior Condition



Not a Competing Objective

Admissibility is not just another loss term to add to the pile. It is the logical prerequisite for any planning goal.

The Necessary Foundation

If distinct futures cannot be distinguished, planning inherently fails—regardless of the intelligence of the downstream agent.

The True North

The ultimate goal of representation learning is not matching reality's pixels, but preserving its possibilities.